

PROBLEM ZINDONE

DATA SCIENCE

EXERCISE 3: CASE STATEMENTS

Q1: Classify each product into the three tiers below:

SYNTAX:

SELECT product_name,

price,

CASE

WHEN price > 1000 THEN 'Expensive'

WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'

WHEN price < 100 THEN 'Budget'

END AS price_category

FROM products;

Output

Q2	product_name	price	price_category
	Laptop	1200.00	Expensive
	Phone	800.00	Mid-range
	Keyboard	45.00	Budget
	Monitor	300.00	Mid-range
	Mouse	25.00	Budget

Q2. Label each order by its value

SYNTAX

```
SELECT customer_name,
```

```
amount,
```

```
CASE
```

```
    WHEN amount >= 1000 THEN 'High Value'
```

```
    WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
```

```
    WHEN amount < 500 THEN 'Low Value'
```

```
END AS order_value_category
```

```
FROM orders;
```

Output

customer_name	amount	order_value_category
Alice	150.00	Low Value
Bob	560.00	Medium Value
Charlie	999.99	High Value
Diana	45.50	Low Value
Ethan	1200.00	High Value

Q3. Categorize each employee's position level using both department & salary

SYNTAX

```
SELECT emp_name,
```

```
department,
```

```
salary,
```

```
CASE
```

```
    WHEN department = 'IT' AND salary > 8000 THEN 'Senior IT'
```

```
    WHEN department = 'HR' AND salary > 5500 THEN 'Experienced HR'
```

```
    ELSE 'Staff'
```

```
FROM employees;
```


D3 continued -

Output

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Q4 Assign each student a letter grade based on their score

SYNTAX

SELECT student_name,

score,
CASE
~~grade~~

WHEN score ≥ 90 THEN 'A'

WHEN score BETWEEN 80 AND 89 THEN 'B'

WHEN score BETWEEN 70 AND 79 THEN 'C'

WHEN score BETWEEN 60 AND 69 THEN 'D'

WHEN score < 60 THEN 'F'

Output

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

Q5 Label delivery performance based on the time taken

SYNTAX

```
SELECT delivery_id,  
       delivery_time_minutes,  
       CASE  
         WHEN delivery_time_minutes ≤ 30 mins THEN 'Fast'  
         WHEN delivery_time_minutes BETWEEN 31 AND 60 mins THEN 'On Time'  
         WHEN delivery_time_minutes > 60 mins THEN 'Late'  
       END AS performance  
FROM deliveries;
```

Output

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Q6. Convert the numeric priority into a readable label

SYNTAX

```
SELECT issue_type,  
       priority,  
       CASE  
         WHEN priority = 3 THEN 'High'  
         WHEN priority = 2 THEN 'Medium'  
         WHEN priority = 1 THEN 'Low'  
       END AS priority_label  
FROM tickets;
```


Q6 continued - - -

Output

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

Q7. Calculate attendance percentage and classify result

SYNTAX

```
SELECT student_id,  
       (days_present * 100) / total_days AS attendance_percentage,  
       CASE  
         WHEN attendance_percentage ≥ 90 THEN 'Excellent'  
         WHEN attendance_percentage BETWEEN 75 AND 89 THEN 'Good'  
         WHEN attendance_percentage < 75 THEN 'Needs Improvement'  
       END AS attendance_status
```

Output

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Needs Improvement
3	96	Excellent
4	50	Needs Improvement
5	100	Excellent

Q8. Label the stock status of each product

SYNTAX

```
SELECT product_id,  
       stock_qty,  
       CASE
```


Q8 Continued - -

WHEN stock_qty = 0 THEN 'Out of stock'

WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low stock'

WHEN stock_qty > 5 THEN 'In stock'

END AS stock_status

FROM products_inventory;

Output

product_id	stock_qty	status
1	5	In stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

Q9. Classify each class by the number of enrolled students

SYNTAX

SELECT subject,
enrolled_students

CASE

WHEN enrolled_students ≥ 25 THEN 'Large'

WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'

WHEN enrolled_students < 10 THEN 'Small'

END AS class_size_category

FROM classes;

Output.

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

Q10. Apply a discount flag based on the payment method and amount

SYNTAX

```
SELECT payment_id,  
       payment_method,  
       amount,
```

CASE

WHEN payment_method = 'Cash' AND amount $\geq$ 200 THEN 'Eligible for Discount'

Else 'Not Eligible'

FROM payments;

Output

payment_id	payment_method	amount	discount eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	Paypal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount